Probably Approximately Correct String Theory: A Verification Discipline for the Landscape

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Abstract

The string landscape contains on the order of 10^{500} vacua, and no experiment distinguishes most of them, so "unified string theory" is two conflated problems—*verification* (does some vacuum match our universe?) and *selection* (why that one?). We separate them, and build a *verification discipline* for the first: a probably-approximately-correct certificate that bounds a candidate's error even when the verifier is itself approximate (Theorem 3), a proof that search over the flux superpotential is *not evolvable* (Denef–Douglas uniformity + Valiant's contrapositive, measured on 380 real SUSY vacua), a *measured* log-uniform prior for $|W_0|$ (CDF linearity $R^2 = 0.92$), a certified *Mitchell Verification Deficit* that is total in the Standard-Model sector but *already climbed to* $m = 5$ in the topology and flux sectors (CYTools reproduces the quintic's $h^{1,1}, h^{2,1}, c_2, \kappa$; JAXVacua reproduces the D3 tadpole as the exact symplectic pairing), and a *boundary map*—a taxonomy of the four senses in which the problem is "uncomputable," each with its certified positive remainder. We do not claim to select the vacuum. We claim something narrower and we believe more useful: *the field currently has no verification layer for its unverifiable theory, and this is one.*

1 Introduction

String theory is not a single theory but a family of $\sim 10^{500}$ metastable vacua—the *landscape* [1, 2, 3]. "Unified string theory" decomposes into two claims the literature habitually fuses: **verification** (there exists a vacuum matching our universe) and **selection** (a measure singles it out). The first is checkable in principle; the second is the open measure problem. This paper concerns the first, and does so rigorously.

Two facts make verification hard. First, the search space is astronomically large. Second, and more fundamentally, the *verifier is not ground truth*: the map from a compactification to its low-energy observables is computed by an approximate solver, with systematic error from α' corrections, non-perturbative truncations, and the Kaluza–Klein tower. Any rigorous statement about verification must contend with an *imperfect labeler*.

We import the framework built for exactly this regime—Valiant's probably-approximately-correct (PAC) learning [4]—and apply it to verification itself. The result is a certificate whose two error terms are separately estimable, three consequences (a verify/search separation, a necessary-condition conjecture, and a measurable-physics bound), and, on running the machinery against real data, a set of empirical findings summarized in the abstract.

Field context. Independently of us, the field is converging on verification from three directions: Michael Douglas is formalizing QFT in Lean 4 [13]; PhysProver is the first LLM theorem-prover for physics [14]; and a 2025 line formalizes *falsification* targets at a collider [15]. Our contribution is the complementary angle: guards, certificates, and measured limits—verification for the regime where ground truth is absent.

2 The two-level certificate

Let X be the observable space, D a measure over X, $c^* : X \to Y$ the target (our universe's true physics), and $h : X \to Y$ a candidate's prediction. We observe c^* only through an approximate **oracle** $\widehat{O}$. Define

$$g(x) = |h(x) - \widehat{O}(x)|, \qquad e(x) = |\widehat{O}(x) - c^*(x)|,$$

and the true error

$$\mathrm{err}_D(h) = \Pr_{x\sim D}\big[\,|h(x) - c^*(x)| > \varepsilon\,\big].$$

The triangle inequality and a union bound give, for any $\beta \in (0, \varepsilon)$,

$$\mathrm{err}_D(h) \;\le\; p_g(\varepsilon - \beta) + p_e(\beta), \tag{1}$$

where $p_g(\tau) = \Pr[g > \tau]$ and $p_e(\beta) = \Pr[e > \beta]$. The two terms are *separately estimable*: p_g compares two things we can both compute (h and $\widehat{O}$), while p_e is the oracle's own failure rate, estimated by validating $\widehat{O}$ on points where c^* is known.

Theorem 1 (Candidate bound)**.** *For fixed h, with n samples $x_1, \ldots, x_n \sim D$ and empirical disagreement rate $\hat{p}_g$, with probability $\ge 1 - \delta_s$,*

$$p_g(\varepsilon - \beta) \;\le\; \hat{p}_g + \sqrt{\frac{\ln(1/\delta_s)}{2n}}.$$

Theorem 2 (Oracle bound)**.** *On m points where c^* is known, with empirical exceedance $\hat{p}_e$, with probability $\ge 1 - \delta_e$,*

$$p_e(\beta) \;\le\; \hat{p}_e + \sqrt{\frac{\ln(1/\delta_e)}{2m}}.$$

Theorem 3 (Two-level Valiant certificate)**.** *With probability $\ge 1 - \delta_s - \delta_e$,*

$$\mathrm{err}_D(h) \;\le\; \hat{p}_g + \hat{p}_e + \sqrt{\frac{\ln(1/\delta_s)}{2n}} + \sqrt{\frac{\ln(1/\delta_e)}{2m}}.$$

The certificate is the tuple $(\varepsilon, \beta, n, m, \hat{p}_g, \hat{p}_e, \delta_s, \delta_e)$. This is verification in the presence of an imperfect labeler—the generalization of PAC's clean oracle that string theory actually requires.

3 Verification without learnability; search with it

Theorem 3 never invokes the complexity of the hypothesis class: *verifying what you are handed is cheap*, a Bernoulli-rate estimate. *Search* is different—it selects the empirically best of many hypotheses, which is biased, and bounding the bias needs uniform convergence (VC dimension, or the statistical-query dimension SQdim). The consequence is a graceful-degradation statement: if search is infeasible, the verifier survives and can still certify any externally proposed candidate.

4 The measurable-physics bound, and the total deficit

Make D **finite**: X is a finite list of N log-space observables. Then the candidate term is *exact*—no sampling:

$$p_g(\varepsilon - \beta) \;=\; \sum_{i=1}^{N} w_i\, \mathbf{1}\big[\,|\ln(h(x_i)/\widehat{O}(x_i))| > \varepsilon - \beta\,\big],$$

and the certificate collapses to the oracle term alone, bounded by m, the number of *validated oracle outputs*—observables that are simultaneously *measured* (c^* known), *computable* ($\widehat{O}$ evaluates them), and *validated* ($\widehat{O}$ calibrated against c^*).

The honest finding. For our universe, c^* is known but $\widehat{O}$ is not (the compactification is unknown); for exact/tractable backgrounds, $\widehat{O}$ is computable but they are not our universe. The intersection—the true m—is **zero** today. With $m = 0$ the radius $\sqrt{\ln(1/\delta_e)/(2m)}$ is **infinite**: the certificate is vacuous, and the verification deficit the *Mitchell Verification Deficit* is *total*, not "$\sim 31\%$" as an earlier conflation of "measured" with "validated" suggested. The binding constraint is validated oracle outputs, not compute; each observable that gains a calibrated error model moves m from 0 toward the $\sim$ 10–19 targets and shrinks the radius from ∞.

The deficit is a working instrument, not a slogan. We have already moved m by building and validating oracles, sector by sector. The CYTools topology oracle reproduces the quintic's four independent invariants—$h^{1,1} = 1$, $h^{2,1} = 101$, the second Chern class $c_2 = [-250, 50, 50, 50, 50, 50]$, and the Kähler-basis triple intersection $\kappa = 5$—exactly ($m = 4$). The JAXVacua flux oracle reproduces the D3 tadpole as the exact symplectic pairing $N_{\text{flux}} = F^\top JH$ with $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, recovered at machine precision ($r = 1.0$, residual $\sim 10^{-14}$) from 107 real SUSY vacua ($m = 1$). Total: $m = 5$ — the topology and flux sectors each now carry a calibrated oracle, and the Standard-Model sector remains the open rung ($m = 0$). This is the predicted closure, one validated observable at a time, and it is how the deficit is to be read: as a measurement of how much verification a theory has—which this work has begun to supply.

5 The negative result: W_0 is log-uniform, hence not evolvable

Denef–Douglas [6] shows the flux superpotential W_0 is statistically uniform in the complex plane across the flux ensemble, so the map flux $\to W_0$ carries $\sim$zero mutual information. We measured the distribution directly.

Measurement. 380 SUSY flux vacua, sampled by JAXVacua's natural (ISD) measure [11]. The empirical CDF of $\log|W_0|$ is linear to $R^2 = 0.92$ (the $|W_0|^2$ CDF is not, $R^2 = 0.45$):

variable	CDF linearity R^2
$\log\|W_0\|$ (log-uniform)	**0.923**
$\|W_0\|^2$ (area-uniform)	0.450

So $|W_0|$ is **log-uniform**—the standard KKLT prior, now measured rather than assumed. (Uniform-in-a-box sampling does *not* reproduce this: $R^2 \approx 0.2$–0.3 regardless of box width; the natural measure is required.)

Valiant's evolvability theorem [5]: *evolvable* $\Rightarrow$ *learnable*. A log-uniform (hence zero-information) map is not learnable, hence—by the contrapositive—**not evolvable**: no certificate-only evolutionary search over W_0 can converge. The graceful-degradation statement of Section 3 applies: the *search dies, the verifier survives.*

A direct consequence, measured: $\Pr(|W_0| < 1) = 42.6\%$ (so "small W_0" eliminates $\sim 57\%$, not the $\sim 95\%$ that biased sampling suggested), while the KKLT target $|W_0| \sim 10^{-4}$ lies 7.6 nats below the observed range—its tuning probability is an extrapolation the data cannot confirm.

6 Empirical characterization: structure, constraints, uniformity

Structure mining. A non-linear miner (mutual information with permutation test, plus a ridge-on-quadratic-features learnability test, self-validated on the deterministic tadpole control at $r = 0.99$) finds no strong geometric fingerprint in the flux-sector observables: W_0 random (ridge $r = 0.10$), g_s weakly structured ($r = 0.21$), the gauge couplings at noise ($r \leq 0.17$). The geometry-pinning structure, if it exists, lives in the observables we have not yet computed (the full spectrum with instanton corrections).

Constraint verification. Treating constraints as certifiable statements—"this constraint eliminates fraction p of the sampled vacua, with confidence $1-\delta$"—we measure, on the natural-measure sample:

constraint	eliminates	95% CI
weak coupling ($g_s < 1$)	68.7%	[59%, 78%]
small W_0 ($\lvert W_0\rvert < 1$)	57.4%	(measured)

A single constraint eliminates a large, certified fraction; a chain of them narrows by elimination, which is the search-free route the framework is designed for.

7 The boundary of the computable

The object we actually want—pin down the compactification—is uncomputable in several distinct senses. The contribution of this paper is to *name* them and certify each as data, so that "uncomputable" is a map, not a wall:

sense	meaning	the move
intractable	answer exists, no feasible algorithm	prove intractability
unmeasurable	answer exists, data/ground-truth absent	compute the deficit
underdetermined	no unique answer exists	compute the set's invariants
undecidable	provably no algorithm	prove undecidability

Seven questions are certified to date (search over W_0 = intractable; verify-today and KKLT-tail and full-EFT and small-Λ = unmeasurable; the measure and the inverse problem = underdetermined). **None is undecidable**—the landscape problem is an *epistemic* and *underdetermination* limit, not a Turing limit. That is itself a finding: the path forward is not "invent a faster computer" but "build the missing oracles" and "compute the invariants of the underdetermined set."

8 A worked failure, and what the format bought

The strongest evidence for the gate format is that it found one of its own certificates wrong.

A gate of the form *"is the textbook constant the right reference?"* was recorded with every required field — name, result, evidence, producer, falsifier, remedy — and graded VERIFIED. Its *result* field stated that a reference constant was the asymptotic value for the ensemble being graded, and that comparing against it measured finite-sample bias. The constant was 0.5307; the graded system was unitary, and the corresponding value under the same estimator is 0.6136. The constant is the real-symmetric (GOE) one. The recorded reason was wrong, and it had propagated from a module name into a docstring, two documents and a certificate.

What the format did, in order: (i) the falsifier was a checkable sentence — *"comparing against the constant would report a 4σ failure of a correct computation"* — and it was checked, the failure occurred, so one of the two numbers had to be wrong; (ii) the evidence was a computation rather than a citation, so the control could be re-run at the right ensemble and size, where it disagreed; (iii) the remedy field already said the right thing, which is what made the error legible: the practice was sound and the *reason* was not.

The correction renamed the constant for its ensemble, added the unitary reference measured from our own sampler rather than quoted, kept the old name as a deprecated alias — a constant that silently changes value is worse than one with a wrong name — and added a test that fails in both directions. The previous test asserted only that the control was more than 0.05 from the constant, which is true under either ensemble and could therefore never catch the defect.

The same week produced a second, duller instance of the same point: four documents — including the submission package itself and the file the package quotes for paper 2's numbers — were committed with git conflict markers in them. Every numeric claim in them was correct; the files were unreadable. The suite had a dozen checks on numbers and none on documents, so the guard is now one that walks every committed text file and fails on a marker. *The check you do not have is the one that ships the defect.*

A verification system that has never found itself wrong is not verification.

9 Discussion and limits

The framework does not unify. It does not select a vacuum; the measure problem remains open physics. It does not eliminate oracle error; it bounds it. Every certificate is conditional on the stipulated observable measure D.

What we do not claim. No new prediction, no measure, no unique compactification. The negative result is scoped to the IIB flux corner and to W_0 (g_s is weakly structured; the full-spectrum observables are unmeasured). The log-uniform prior is measured over 3.7 nats; its extension to the KKLT tail is explicitly an extrapolation.

What we do claim. A verification discipline for a theory that currently has none: a certified deficit, a mutation-tested computation vault, a measured prior, a negative result, and a boundary map—built in the exact gap the field's leaders (Douglas [13], PhysProver [14]) are independently formalizing. The two approaches—our guard/certificate discipline and their Lean formalization—are complementary, not competing.

10 Conclusion

The obstacle to unified string theory is a shortage of *checkable claims*, and the deepest checkable claim in reach is not "is this vacuum ours?" but "is the compactification→observables map learnable?"—answered negatively for W_0, and mapped for the rest. Verification, done with Valiant's guarantees, is the honest first move; the boundary map makes the *second* move (where does computation end?) equally honest. We have not solved the theory. We have built, tested, and measured the missing verification layer—and the field's convergence on the same gap suggests that is the right thing to have built.

References

[1] R. Bousso and J. Polchinski, *Quantization of four-form fluxes and dynamical neutralization of the cosmological constant*, JHEP (2000).

[2] M. R. Douglas, *The statistics of string/M theory vacua*, JHEP (2003).

[3] L. Susskind, *The anthropic landscape of string theory* (2003).

[4] L. G. Valiant, *A theory of the learnable*, CACM 27(11), 1984.

[5] L. G. Valiant, *Evolvability*, J. ACM 56(1), 2009.

[6] F. Denef and M. R. Douglas, *Distributions of flux vacua*, arXiv:hep-th/0404116.

[7] M. Kearns, *Efficient noise-tolerant learning from statistical queries*, J. ACM 45(6), 1998.

[8] V. Feldman, *A complete characterization of statistical query learning with applications to evolvability*, FOCS 2009.

[9] L. G. Valiant, *Probably Approximately Correct*, Basic Books, 2013.

[10] M. Demirtas et al., *CYTools: A Software Package for Analyzing Calabi-Yau Manifolds*, arXiv:2211.03823.

[11] A. Dubey, S. Krippendorf, A. Schachner, *JAXVacua—a framework for sampling string vacua*, arXiv:2306.06160.

[12] S. Kachru, R. Kallosh, A. Linde, S. Trivedi, *de Sitter vacua in string theory*, arXiv:hep-th/0301240.

[13] M. Douglas et al., *Formalizing QFT in Lean 4*, arXiv:2603.15770.

[14] *PhysProver*, arXiv:2601.15737.

[15] *How to falsify string theory at a collider*, arXiv:2412.13192, PRR 2025.